

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1496

San Juan 138 kV
February 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1496. The Interconnection Customer proposes a 108.3 MW battery energy storage facility interconnecting at the San Juan 138 kV substation in the West Virginia area. The request was received on June 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 438 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,781,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Panda – Goat Rock 138 KV Line	Panda – Northern Steel Casting 138 KV	122.1	188	9.9
Goa Rock – Northern Steel Casting 138 KV	Panda – Northern Steel Casting Substation 138 KV	118.0	199	9.9
Panda Substation – Northern Steel Casting 138 KV	Northern Steel Casting – Goat Rock 138 KV	114.8	226	11.0
Menlo – Carlsbad Waterfield 138 KV	Menlo – W Woodland 138 KV	108.7	205	9.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Panda – Goat Rock Substation 138 KV	Rebuild line with higher-capacity conductor	\$9,160,000	32% (\$2,945,000)
Northern Steel Casting – Goat Rock 138 KV	Rebuild line with higher-capacity conductor	\$7,619,000	32% (\$2,450,000)
Panda – Northern Steel Casting 138 KV	Rebuild line with higher-capacity conductor	\$3,440,000	32% (\$1,106,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Menlo – Carlsbad Waterfield 138 KV	Reconductor line	\$1,690,000	76% (\$1,280,000)
	Total allocated to project		\$7,781,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.